The Generation Structure of Fermions: Particles as Completed Closure Cycles in a Sea of Light

Spontaneous Differentiation of Forces from the Universal Interaction, Spontaneous Differentiation of Particle Species by Parity, Address, Coherence, and Hair, and the Counting of Abundances

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Position: The Dimension Generation series, main paper 9. A synthesis of the trilogy (two-grammar decomposition [3], counting readouts [4], lock dynamics [5]) and the Paper-D preliminary experiment suite [9]. **This paper is derivation and synthesis only, and contains no new numerical experiments and no new figures.** All quantitative facts rest on published papers and committed reproduction packages, and every one of them can be traced one-to-one through the source correspondence (Section 6).

Claims — The Mechanism by Which Particles Spontaneously Differentiate from the Form of the Wave

The key is the inversion of default and exception.

The default state is a sea of photons. Zero-square-sum closure (the fully coherent shared mode) is the axiom itself. Nothing needs to be "made" — the raw material of the system is lightlike from the start.

The first branching is parity. Odd harmonics = fermionic; even harmonics = bosonic.

The second branching is address capture. Within the coherent sea, only those chains of exchange that happen to complete the closure cycle of a register become recurring closures — that is, particles.

Matter and antimatter are given in pairs by the geometry of cycle completion. The halfway point of the cycle is $\lambda^{62} = -1$ (conjugate, opposite sign), and the full circuit is $\lambda^{124} = +1$ — the sign of the double cover itself gives the pair.

In other words, particles are rare events that completed the closure cycle in the sea of light. They are "almost absent" because completing the closure cycle is difficult, and "not zero" because the resolution is finite and the completion probability is strictly positive — both emerge automatically as structure.

What follows is nothing more than the derivation that led to this claim.

Position of This Paper

The paper-9 series has sought a mechanism by which both the species of particles and the kinds of forces differentiate spontaneously from the form of the wave alone, without either being given as input. The trilogy established, at machine precision, the two-term identity of the universal interaction and the unique differentiation of the two grammars [3], the structure of counting readouts [4], and the quantization mechanism together with the elimination approach to selection [5]; the Paper-D preliminary experiment suite [9] measured the register relation of the two alpha roots, clock blindness, aliasing, and the mass beat. This paper synthesizes these into a single generation structure. Every proof is either (i) a citation of a published measurement or (ii) an elementary identity in the main text; no new experiment is performed.

1. The Problem

Three points must be explained. (i) Where do the species of particles come from — without an external classification table. (ii) Where do the kinds of forces come from — without inputting coupling rules for each particle species. (iii) Why is the abundance of matter "extremely small yet strictly nonzero" — without fine-tuning parameters.

2. Derivation I — The Universal Interaction: There Is Only One Force

The interaction present in the system is one and only one, the real-orthogonal exchange collision, and the transfer in a single collision decomposes identically into the two terms

$$dN_B = \underbrace{\sin^2\theta\,(N_A - N_B)}_{\text{magnitude term}} + \underbrace{\sin 2\theta\,\mathrm{Re}\langle a|b\rangle}_{\text{overlap term}}$$

and **no third term exists** [3]. This identity does not ask what the partner is. Hence this is the universal interaction, and the distinction between forces is not a distinction between interactions but **a distinction of which term the partner's waveform activates**:

- The magnitude term acts on everything that has a norm — the gravity type (universal, additive, unscreenable, no quantization) [3]
- The overlap term acts in the charge type only on pairs that share harmonics and carry hair — the Coulomb type (amplitude product, ±, two neutralization-screening mechanisms, rational quantization) [3]
- A fully shared relation carrying no hair bears neither charge and mediates both — the photon (Section 3). Note that the two "absences" stem from separate axes: the absence of mass follows from full sharing ($\det\Gamma = 0$, Section 3, axis (3)), and the absence of charge-type flow follows from the absence of hair (Section 3, axis (4))

This assignment is not a similarity of properties but a unique correspondence that excludes the reverse assignment (Section 6.5 of [3]). That is, **the kind of force is an alias of the kind of particle, and the kind of particle is an alias of the form of the wave.**

3. Derivation II — The Four-Axis Decision Tree: Species Differentiate from the Form of the Wave

The species of a particle is determined solely by four independent properties of the form of the wave.

Axis (1) Parity (statistics). Odd harmonics are antiperiodic, $f(\theta + \pi) = -f(\theta)$, and two-valued — fermionic, requiring a double cover. Even harmonics are single-valued — bosonic. The fermionic construction from equal-amplitude bundles of odd harmonics and their collisions have been measured [8].

Axis (2) Binding (address and quantization). A family of harmonics bundled in integer ratios forms a harmonic family = a closure, and recurs as an exact finite-order root of the exchange operator, $U^n = \pm I$ [2]. A closure carries a coprime address m/n on the register n, and **the address gives the charge value**: the elementary charge is $1 - R = \sin^2(23\pi/124) = 0.302822$, agreeing with the dimensionless elementary charge $\sqrt{4\pi\alpha}$ to 0.16 ppm [2][3]. The effective charge at high energy is a reading of the same elementary charge on a fivefold-refined register ($620 = 5 \times 124$, clock $1240 = 5 \times 248$); to an observer with a coarse clock, the fine root is read as exactly folded back onto the electron's address (aliasing = a candidate mechanism for charge universality) [9]. Relations that are not bound remain in the sea.

Axis (3) Coherence (mass). For a pair (a, b), the Gram invariant

$$\det\Gamma = N_A N_B - |\langle a|b\rangle|^2$$

satisfies $\det\Gamma \geq 0$ by Cauchy–Schwarz, with equality if and only if $b \propto a$ (full sharing) — this is an identity, not an experiment. Reading $\det\Gamma$ as the mass-squared-type invariant, **mass is incoherence, and for full sharing the reading called mass can in principle never arise**.

Axis (4) Hair (the carrier of charge). Charge-type flow arises only in shared pairs carrying ± carriers (hair), and neutralization occurs through two mechanisms: zero components and channel non-sharing [3].

From the four axes the archetypes emerge:

- **The photon** = a fully shared even-difference mode carrying no hair. It is the quantum of the shared-channel selection rule $\Delta k = \pm 2$ (measured, coupling 0.5 [3]) — the mode of the difference of two odd harmonics = even; a boson because it is even (axis (1)), massless because it is fully shared (the identity of axis (3)), chargeless because it carries no hair (axis (4)), free because it holds no address. The four properties each follow from independent axes. The long-range inverse square is borne by the mediation of harmonic closure [7]
- **The charged fermion (electron type)** = odd, bound (an address on register 124), incoherent bundle, shared with hair
- **The neutral fermion** = odd, bound, hair-free or non-shared (subject to gravity only: the magnitude term reads the norm regardless of address [3])
- **The baryon** = a bound closure topologically protected by winding number (baryon number = winding number; stability = the absence of a decay destination). Within this framework this identification is at the stage of a **correspondence hypothesis** (externally, the Skyrme lineage [12][13] demonstrates an isomorphic structure), and its derivation is placed in Open Problem 4

One vacant slot remains explicit in the decision tree: **even, bound, with hair** (a charged massive boson — the slot for W±) — Open Problem 2.

4. Derivation III — Matter and Antimatter: The Geometry of Cycle Completion Counts in Pairs

The eigenvalues of the elementary-charge register satisfy $\lambda^{31} = i,\ \lambda^{62} = -1,\ \lambda^{124} = +1$ [2]. The halfway point of the completed cycle, $\lambda^{62} = -1$, is conjugation and sign reversal, and at the full circuit $\lambda^{124} = +1$ the system returns to itself. Combined with the 2:1 covering of the spinor (the state period is twice the observation period [9]), **the geometry of completing the closure cycle itself counts particle and antiparticle in pairs**. No additional symmetry is needed to produce the pairs.

5. Derivation IV — Abundance: Scarcity and Nonzeroness Emerge Together

A system of resolution N has $M = N(N-1)/2$ relational waves [6]. Only the integer-ratio pairs (pairs in which one is an integer multiple of the other — pairs bundled into a harmonic family, axis (2)) can complete the closure, and their number is

$$\#\{(i,j) : i < j,\ i \mid j\} = \sum_{i=1}^{N} \left(\left\lfloor \frac{N}{i} \right\rfloor - 1 \right) = N \ln N + (2\gamma - 2)N + O(\sqrt{N})$$

(γ: Euler's constant; Dirichlet's divisor-sum estimate). Hence the fraction of relations that can become closures is

$$f(N) = \frac{2\,(\ln N + 2\gamma - 2)}{N} + O(N^{-3/2})$$

— **it falls rapidly as N expands, yet never becomes zero**. Both "matter is almost absent" and "matter strictly exists" emerge automatically from this single fraction. Equating the observed baryon-to-photon ratio $\eta \approx 6 \times 10^{-10}$ with $f(N)$ gives $N \approx 8 \times 10^{10}$ — η **is read as a fossil of the resolution at the instant the inflationary expansion halted in three directions** (the value N itself is a reading from observation, not a derivation — Open Problem 8).

Baryogenesis in standard cosmology takes Sakharov's three conditions and the magnitude of CP violation as external inputs [17]. The claim of this framework is their replacement: **the smallness of the asymmetry is not fine-tuning but the counting of integer-ratio pairs** (the detailed correspondence with B-violating dynamics is left to the Open Problems).

6. Source Correspondence

The verified facts on which each sentence of the Claims relies:

- **The sea of light = zero-square-sum closure**: the axiom system [1]. Full sharing ⇒ no mass is the Cauchy–Schwarz identity in the main text (Section 3, axis (3))

- **The parity branching**: the measured fermionic construction and collisions of odd-harmonic bundles [8], the implementation of $\pm$ hair [3]
- **Address capture, recurrence, quantization**: exact finite-order roots [2], counting readouts and Niven intersections [4], Arnold tongues, observational selection, and clock inheritance [5], the register arithmetic of the two alpha roots, clock blindness, aliasing, and the mass beat [9]
- **The λ periods of matter/antimatter**: Section 6.5 of [2]
- **The accounting of abundance**: the mechanism of relational waves $M = N(N-1)/2$ [6]; the counting is Section 5 of the main text (elementary)
- **The universal interaction and force as an alias of species**: the two-term identity and the unique assignment [3]
- **The photon = mediator**: the null theorem and the $\Delta k = \pm 2$ ladder [3], the inverse-square mediation of harmonic closure [7]

7. Relation to Prior Work — Evidence of Convergence and Differences

The external literature is not the basis of the derivations in this paper. It shows independent realizations of isomorphic configurations and historical precedence.

Particles as stable structures in a medium. Kelvin's vortex atoms [10]: atoms = stable vortex structures in a universal medium — the prototype of "default = medium, particle = exceptional stable structure." Wheeler's geons [11]: electromagnetic waves bundled by their own gravity behaving like particles, "mass without mass" — the first quantitative attempt to build matter from light (difference: requires a background spacetime and gravitational binding). Skyrme [12] and Finkelstein–Rubinstein [13]: baryon number = topological winding number, and spin statistics from the parity of the winding number via the double cover — the closest precedent for this paper's baryon correspondence hypothesis and parity branching (difference: solitons of fields on a background spacetime). Hestenes's zitterbewegung electron [14]: the electron = a lightlike helical closed orbit — the single-particle version of "mass = a bundle of lightlike components" (difference: no mechanism generating species). Volovik [15]: fermionic quasiparticles, gauge fields, and effective gravity co-emerge near the Fermi point of superfluid ^{3}He — a modern realization of the "co-emergence of species and forces" (difference: assumes a condensed-matter substructure, low-energy limit only, no rational address quantization). Wilczek [16]: 99% of the mass of the visible universe comes from the dynamics of massless quarks and gluons — empirical support that "mass from massless constituents" is the actual state of nature.

Harmonics and particle species. Kaluza–Klein [18][19]: the mode number on a compact circle gives charge (integer quantization) and a mass tower — the direct precedent for "harmonic number on a circle = charge" (difference: requires extra spatial dimensions, and the address is integer-only, with no coprime ratio, no parity structure, and no universality mechanism). The RNS construction of string theory [20][21]: particle species = vibration modes of a single object, and moreover periodic (Ramond) / antiperiodic (Neveu–Schwarz) boundary conditions with integer / half-integer modes separate fermions from bosons — the direct precedent for the parity branching (difference: requires a worldsheet, a critical dimension, and worldsheet supersymmetry from outside; this paper uses only the parity of the harmonics of the closed circle themselves). Regge–Chew–Frautschi [22]: families of hadron species = ladders of excitations (difference: phenomenology within the strong interaction). As a historical seed, de Broglie [23]: matter = periodicity.

Contrasting frame. Sakharov [17]: the three conditions for baryogenesis — the standard frame whose replacement this paper's counting claims.

Three things are unique to this paper: (i) statistics from the parity of the harmonics themselves (no supersymmetry), (ii) charge from coprime rational addresses (no extra dimensions), (iii) abundance from the counting of finite resolution (no fine-tuning) — and the fact that the three branch simultaneously from a single axiom system.

8. Open Problems

First tier — gaps remaining directly in the claims of this paper

1. **The address-selection problem (uniqueness of the elementary charge).** Register 124 has $\varphi(124) = 60$ coprime addresses, each giving a different charge value. At this stage the framework has no reason why only one of them is realized — **there could just as well be 60 kinds of charge**.

Eliminated: static counting [4], lock weights (width ratio > 461) [5], fixed points [5], observational selection (inherited into the clock) [5], four simple arithmetic screenings [9]. Remaining candidates: mass balance (norm-sector dynamics, untested) and clock inheritance from the parent closure. The final form is "why is the observation clock commensurable with 248?" [5].

2. **The boson spectrum problem.** The only boson derived is the photon. The reason for the existence of massive bosons and charged bosons (the W± slot = even, bound, with hair) has not been derived; it is the vacant slot in the decision tree.
3. **The mass-dictionary problem (energy hierarchy).** Mass = incoherence is qualitatively established, but the functional form giving physical mass from register and address is underived. The naive candidate (register ratio 5, coupling ratio 25) is inconsistent with the generation ratio $m_\mu/m_e = 206.77$ by the model's own precision standard (rejected [9]). Why the energy hierarchy appears as discrete generations is also underived. **We can state the existence of massive things, but we cannot state a single mass value.**
4. **The baryon identification problem.** Baryon number = winding number is at the stage of a correspondence hypothesis; its derivation within this framework (the definition of the winding number, its conservation, and stability = the absence of a destination) is incomplete.

Second tier — connection problems shared with adjacent papers

5. **The distance dictionary**: the translation $\Delta\theta \leftrightarrow r$ and the Kepler-type closure condition $\omega^2 R''^3 =$ const. (shared with the trilogy [3] and the real-center paper [7]).
6. **Sign translation**: whether the ± of the relative phase maps to spatial attraction or repulsion (already fixed as a prediction that will be decided automatically the moment the dictionary is completed [3]).
7. **Geometric connection**: the eight problems of the realization map from three directions to 3+1-dimensional spacetime are held by the closing section of Part C of the trilogy [5]. This paper does not duplicate them.

Third tier — underived items remaining on the side of the premises

8. **The value of the resolution** N: $f(N)$ is a derivation, but $N \approx 10^{11}$ at the halt of expansion is a reading from the observed ratio η, not a derivation. Why the expansion halts at that N and in three directions is identical to the open problem of [6].
9. **The place of the strong and weak interactions**: the menu of forces in this framework consists only of the two grammars and the mediator; SU(2)- and SU(3)-type structures are untouched.
10. **The value of spin**: the parity $\leftrightarrow$ statistics correspondence is established, but from which discrete feature of the internal modes the value of spin as a transformation factor ($\pm 1/2,\ \pm 1$) is read out is underived.

This table of problems is not an enumeration of the weaknesses of this paper but a guide for further progress of the series.

9. Reproducibility

This paper is derivation and synthesis only and contains no new numerical experiments. All the quantitative facts it relies on are already published: the reproduction packages of the trilogy [3][4][5] (every number in one-to-one correspondence with committed runners, with SHA-256 verification), the reproduction bundle of the finite-order roots [2], the reproduction package of the N-body relational waves [6], the derivations and verifications of harmonic closure and the real center [7], and the Paper-D preliminary experiment suite [9] (in the repository, commits 7beea7ed and d802907a, with predictions written in advance and a full record of refutations and corrections). The only new mathematics used in the main text consists of two elementary facts: the equality condition of Cauchy–Schwarz (Section 3) and Dirichlet's divisor-sum estimate (Section 5).

References

Self-citations

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2. Noriaki Kihara, “Discovery of Finite-Order Resonance in Iterated Exchange Scattering — Identifying the Origin of Peaks near Fine-Structure-Constant Values 137 and 128 with a Reproducible Wave-Packet Model”, Version DOI: 10.5281/zenodo.21421367, Concept DOI: 10.5281/zenodo.21421366, 2026.
3. Noriaki Kihara, “Two-Grammar Decomposition of Interaction in an Anonymous Two-Channel Closed Wave System” (Part I of the trilogy), Version DOI: 10.5281/zenodo.21763996, Concept DOI: 10.5281/zenodo.21763995, 2026.
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Addendum (the evidence structure of this paper). Every quantitative claim of this paper corresponds one-to-one to one of the following: (i) measurements in published papers (verifiable through the reproduction package of each DOI), (ii) committed preliminary experiments in the repository (with predictions written in advance and a full record of refutations and corrections), or (iii) elementary identities in the main text. It contains no new experiments and no new figures because the claim of this paper lies not in new data but in the **arrangement** of established facts — the inversion of default and exception.